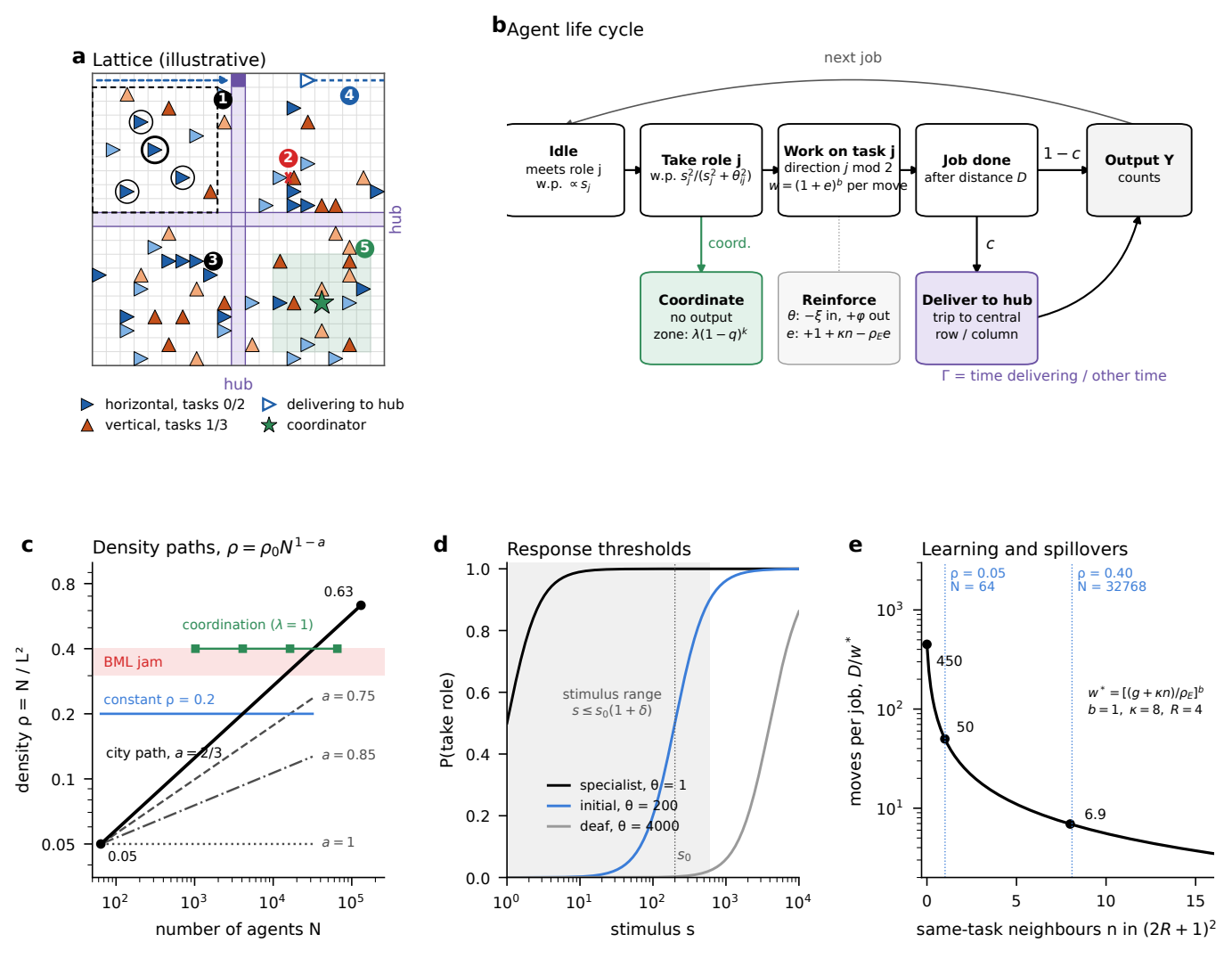
a Lattice (illustrative)
hub
hub
horizontal, tasks 0/2
delivering to hub
vertical, tasks 1/3
coordinator
b Agent life cycle
next job
Idle
meets role j
w.p. ∝ s_j
Take role j
w.p. $s_j^2/(s_j^2+\theta_{ij}^2)$
Work on task j
direction j mod 2
$w=(1+e)^b$ per move
Job done
after distance D
1 − c
Output Y
counts
coord.
c
Coordinate
no output
zone: $\lambda(1-q)^k$
Reinforce
θ: $-\xi$ in, $+\varphi$ out
e: $+1+\kappa n-\rho_E e$
Deliver to hub
trip to central
row / column
Γ = time delivering / other time
c Density paths, $\rho=\rho_0 N^{1-a}$
density ρ = N / L²
number of agents N
0.63
coordination (λ = 1)
BML jam
constant ρ = 0.2
a = 0.75
city path, a = 2/3
a = 0.85
a = 1
0.05
d Response thresholds
P(take role)
stimulus s
stimulus range
$s \le s_0(1+\delta)$
specialist, θ = 1
initial, θ = 200
deaf, θ = 4000
s_0
e Learning and spillovers
moves per job, D/w^*
same-task neighbours n in $(2R+1)^2$
ρ = 0.05
N = 64
ρ = 0.40
N = 32768
450
50
6.9
$w^*=[(g+\kappa n)/\rho_E]^b$
$b=1,\ \kappa=8,\ R=4$